

Training Day 16 Report – Dynamic Analysis & Sandbox Testing

Date: 03 July 2025

Module: Dynamic Malware Analysis

Mode: Practical Lab

Main Focus of the Day

Day 17 focused on **dynamic analysis**—executing malware in a controlled environment and observing its behavior using **monitoring tools and sandboxes**.

Topics Covered

- **Dynamic Analysis Process:**
 - Run malware in VM
 - Observe process, network, file system, and registry behavior
- **Tools Used:**
 - **Process Monitor (Procmon)** – API calls, file/registry access
 - **Process Explorer** – Real-time process tree and DLLs
 - **Wireshark** – Network packet inspection
 - **INetSim** – Fake internet simulation for malware behavior
 - **Cuckoo Sandbox** – Automated dynamic analysis

Practical Activities

- Executed malware sample in an isolated **Windows VM**
- Captured and analyzed process creation, file drops, registry modifications
- Observed DNS and HTTP requests using Wireshark
- Logged and exported behavior report from Cuckoo Sandbox

Key Takeaways

- Dynamic analysis helps reveal malware **behavior, persistence, and C2 communication**
- **Sandboxing** is ideal for automation, but smart malware may detect it
- Tools like Procmon and Wireshark provide **granular insight** into infection trails